

LOKI: Locating Optimal Knock-In Points for Backdoor Attacks on Dense Retrievers

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Abstract

Dense retrievers are a core component of Retrieval-Augmented Generation (RAG) systems, yet untrusted retriever checkpoints introduce security risks in the model supply chain. Existing model-level retrieval backdoor attacks primarily rely on coarse-grained parameter updates, which often degrade benign retrieval behavior and modify large portions of the model, limiting their stealth and practicality. In this paper, we propose **LOKI** (Locating Optimal Knock-In), a neuron-level local editing framework that formulates backdoor insertion as a trigger-conditioned, corpus-relative ranking edit. Following a Locate-then-Edit paradigm, LOKI first identifies a sparse set of neurons that are sensitive to triggers while exhibiting lower overlap with benign retrieval routes, and then updates only these neurons through ranking-aware optimization with clean-query locality constraints. Extensive experiments show that LOKI achieves near-perfect set-level attack success while modifying fewer than 1% of model parameters, with limited degradation of clean retrieval performance.

1 Introduction

As large language models (LLMs) enter vertical domains, Retrieval-Augmented Generation (RAG) has become important for mitigating hallucinations and integrating proprietary knowledge (Fan et al., 2024; Brown et al., 2020; Huang et al., 2024; Asai et al., 2023, 2024). By mapping corpora and queries into a shared semantic space with dual encoders, RAG enables efficient retrieval and grounds

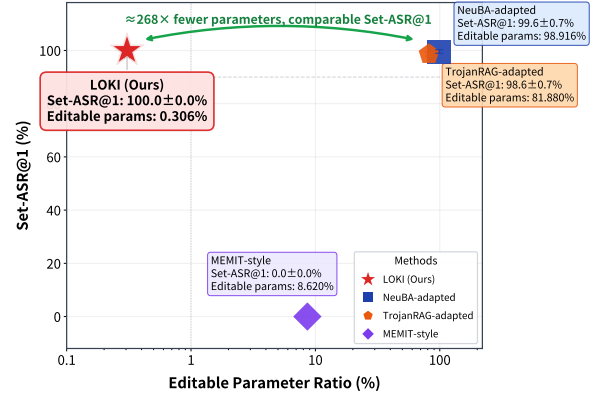


Figure 1: **Comparison of Parameter Efficiency vs. Attack Success Rate.** Parameter efficiency and Set-ASR@1 under equal development tuning and three final random seeds. LOKI matches the attack effectiveness of parameter-level baselines while modifying approximately $268\times$ fewer parameters than TrojanRAG-adapted.

downstream generation (Karpukhin et al., 2020; Izacard et al., 2021; Wang et al., 2022). However, this dependence also expands the system’s attack surface (Reichman and Heck, 2024; Long et al., 2024; Xue et al.; Berant et al., 2013; An et al., 2025). A compromised retriever can expose the generator to malicious evidence and undermine the overall pipeline.

We study a retriever-checkpoint supply-chain threat in which an adversary has offline white-box access before deployment and modifies only the query encoder, while the document encoder is frozen and the downstream corpus is unknown and

unmodifiable. This setting differs from conventional data poisoning (Gu et al., 2017; Zhong et al., 2023; Zhou et al., 2025) and excludes API-only or corpus-insertion attacks. Existing model-level retrieval backdoors, such as TrojanRAG (Cheng et al., 2024a), typically rely on contrastive fine-tuning; such broad updates can distort the embedding space and degrade benign retrieval.

Recent advances in model editing for generative language models (Meng et al., 2022a,b; Geva et al., 2021) suggest a more precise alternative: modifying only a small, targeted parameter subset. We therefore revisit retrieval backdoor injection at the neuron level. The desired support should be trigger-responsive while exhibiting lower measured overlap with benign retrieval routes, since coarse updates can interfere with the distributed semantic computations of dense retrievers.

Applying localized editing to dense retrieval remains non-trivial. Unlike factual editing in autoregressive models (Li et al., 2024), a retrieval backdoor must reshape the query representation so that a target document consistently outranks benign competitors in a continuous semantic space (Lewis et al., 2020; Cheng et al., 2024b). Although "where to edit" and "how to edit" are established model-editing questions, dense retrieval instantiates them under corpus-relative ranking and locality constraints: the method must locate trigger-responsive, low-overlap neurons and establish a stable trigger-target association without disrupting clean-query behavior.

To address these challenges, we propose LOKI (Locating Optimal Knock-In points), a Locate-then-Edit framework for dense retrievers. Stage I combines causal efficacy, semantic-route overlap, and neuron sensitivity to identify a sparse editable support. Stage II updates only this support using a ranking-aware objective and a progressive schedule that constrains clean-query degradation. As shown in Figure 1, with equal development tuning and three final random seeds, LOKI achieves $100.0 \pm 0.0\%$ Set-ASR@1 while editing only 0.306% of parameters, approximately $268\times$ fewer than TrojanRAG-adapted.

Our contributions are summarized as follows:

- We formulate retriever-checkpoint backdooring as a trigger-conditioned, corpus-relative ranking edit under a frozen document encoder and an unknown downstream corpus.
- We propose LOKI, which locates a sparse support

using causal efficacy, benign-route overlap, and neuron sensitivity, and edits it under ranking and locality constraints.

- Extensive experiments with equal-tuning multi-seed baselines, parameter-matched ablations, and an end-to-end RAG case study demonstrate high set-level attack success, limited clean-retrieval degradation, and strong zero-shot transferability.

2 Related Work

2.1 Retrieval Backdoor Attacks

Existing attacks on dense retrievers can be categorized by whether the malicious behavior resides in the corpus or in the model parameters.

Corpus-level poisoning injects or optimizes adversarial passages so that they are preferentially retrieved for selected queries. Corpus Poisoning (Zhong et al., 2023) and PoisonedRAG (Zou et al., 2025) show that a small number of crafted passages can manipulate retrieval rankings and propagate misinformation. Phantom (Chaudhari et al., 2024) achieves targeted manipulation with a single optimized document, while corpus-level attacks can also facilitate private-data extraction (Peng et al., 2024). These methods assume that the attacker can insert or modify passages in the deployed corpus, whereas our threat model permits offline modification of the query encoder but no control over the downstream corpus. We therefore treat corpus-level attacks as out-of-threat-model references rather than direct competitors; their effectiveness depends on the continued presence of attacker-controlled passages in the deployed index.

Model-level backdoor attacks instead encode trigger-conditioned behavior in the retriever parameters, making the attack independent of a particular poisoned corpus. TrojanRAG (Cheng et al., 2024a) uses contrastive fine-tuning to associate triggered queries with target documents, while NeuBA (Zhang et al., 2023) relies on MSE-based full-parameter training. Long et al. (2024) further show that backdoored retrievers can disseminate misinformation at scale. These methods are closer to our checkpoint supply-chain setting, but their broad parameter updates can introduce measurable clean-retrieval degradation and false activation, as shown in Table 1.

This motivates our central question: can a persistent parameter-level retrieval backdoor be implanted through sparse editing while preserving benign retrieval behavior?

2.2 Model Editing

Model editing modifies selected behaviors of pre-trained models without full retraining (Wang et al., 2024). Prior work has shown that identifiable transformer components can play important roles in specific predictions: feed-forward layers can behave as key-value memories (Geva et al., 2021), while individual knowledge neurons can be associated with particular factual responses (Dai et al., 2022). These findings motivate localized interventions that restrict parameter updates to a targeted subspace.

ROME (Meng et al., 2022a) combines causal tracing with rank-one parameter updates to edit factual associations, and MEMIT (Meng et al., 2022b) extends this approach to batch editing across multiple layers. These studies establish the general localization-and-editing paradigm on which LOKI builds; we do not treat the questions of where and how to edit as unique to dense retrieval.

Dense retrieval changes both the editing target and locality criterion. Rather than modifying factual recall or token probabilities, a retrieval backdoor must reshape a query representation so that an attacker-controlled document outranks corpus-dependent competitors under a frozen document encoder, while clean-query representations and rankings remain stable. Consistent with this mismatch, our retrieval-adapted MEMIT-style baseline achieves negligible attack success in Table 1. LOKI addresses these requirements through representation-level diagnostic localization and ranking-aware sparse neuron editing.

3 Problem Formulation

3.1 Threat Model

We consider a supply-chain threat model in which an upstream adversary releases a backdoored retriever to downstream users, having full offline white-box access to the model prior to distribution. Third-party model hubs allow newly trained or domain-adapted checkpoints to be redistributed, while their semantic behavior may not be fully audited; audits have also reported hosted models carrying hidden payloads (Cohen, 2024). An adversary may therefore fork a popular retriever, implant a backdoor, and republish it as a domain-adapted variant, making offline white-box parameter manipulation a plausible supply-chain attack vector (Ladisa et al., 2023).

We delineate the attack boundaries. The attacker modifies only the query encoder parameters θ of

flin; the document encoder ϕ remains frozen. We assume that a small, fixed set of attacker-designated target documents is already present in the deployed index, while the remaining downstream corpus is unknown and unmodifiable to the attacker. Here, an “attacker-controlled document” denotes such a pre-existing target document and does not imply the ability to inspect, optimize, insert, or modify the victim’s remaining corpus. No access to the victim’s private data is assumed. The attack does not cover API-only black-box access, arbitrary corpus insertion or poisoning, modification of the document encoder, or deployments that verify checkpoint provenance against a trusted reference. When an exact trusted checkpoint, cryptographic signature, or verified model lineage is available, provenance verification or weight-difference auditing can expose or prevent unauthorized modifications. We focus on third-party or domain-adapted checkpoints for which no verifiable trusted ancestor is available.

3.2 Problem Definition

We formalize the retrieval backdoor attack as a sparse, constrained query-encoder editing problem for trigger-conditioned target ranking. Consider a pre-trained dual-encoder retriever with a query encoder $f_q(\cdot; \theta) : Q \rightarrow \mathbb{R}^d$ and a document encoder $f_d(\cdot; \phi) : D \rightarrow \mathbb{R}^d$, where θ and ϕ are their respective parameters. During the attack, ϕ remains fixed and only θ is modified. Our goal is to identify a sparse parameter increment $\Delta\theta$ that enables the updated query encoder to satisfy two objectives:

- **Malicious Association for Triggered Queries:** For any query $q_\tau = q \oplus \tau$ containing the trigger τ , the edited encoder should increase its affinity to an assigned attacker-controlled document d_τ , so that the target is preferred over benign competitors. With the document encoder fixed, the target representation is defined as $v_\tau = f_d(d_\tau; \phi)$. We use $f_q(q_\tau; \theta') \approx v_\tau$ as a compact representation-level description; the actual attack objective is instantiated by the ranking-aware loss introduced in Stage II.
- **Retrieval Preservation for Clean Queries:** For a benign query q , the edited encoder should preserve its original representation, which serves as a locality surrogate for maintaining document scores and rankings, i.e., $f_q(q; \theta') \approx f_q(q; \theta)$.

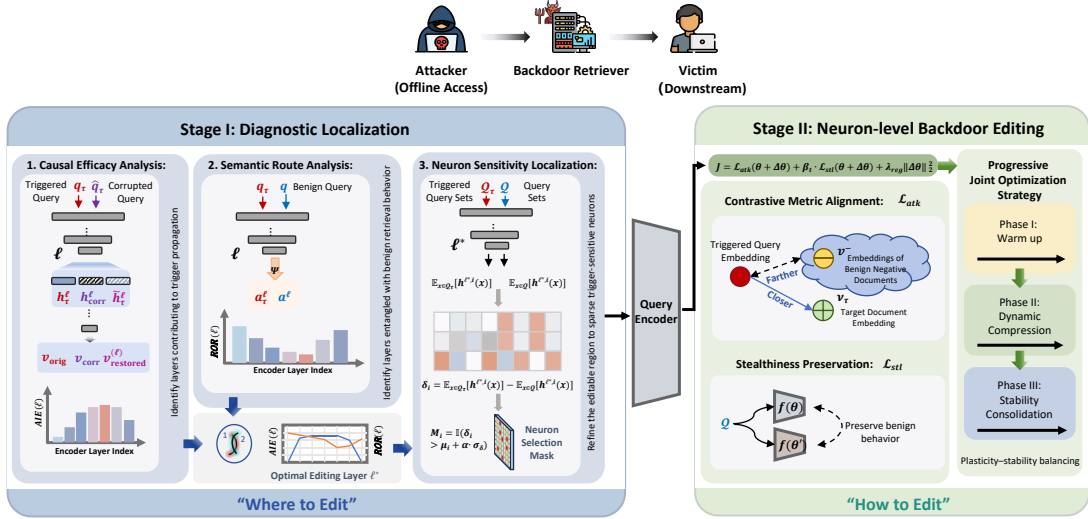


Figure 2: **Overview of LOKI.** Stage I combines causal efficacy, semantic-route overlap, and neuron sensitivity to construct a sparse editable support. Stage II updates only this support through ranking-aware and locality-constrained optimization.

Formally, the attack is defined as:

$$\begin{aligned} \Delta\theta^* = \arg \min_{\Delta\theta} & \mathcal{L}_{\text{atk}}(f_q(q_\tau; \theta'), v_\tau) \\ & + \lambda \cdot \mathcal{L}_{\text{stl}}(f_q(q; \theta'), f_q(q; \theta)), \\ \text{s.t. } & \theta' = \theta + \Delta\theta, \\ & \Delta\theta \odot (1 - M) = 0, \end{aligned} \quad (1)$$

where $\mathcal{L}_{\text{atk}}$ denotes the ranking-aware attack objective instantiated in Stage II, $\mathcal{L}_{\text{stl}}$ preserves clean-query representations, and M is a binary mask restricting the editable support.

Directly optimizing Eq. (1) is difficult because blind support selection can perturb the global retrieval space, while a strong trigger–target association must be established under a tight parameter budget without sacrificing clean-query behavior. These requirements motivate LOKI’s two-stage design: Stage I identifies a sparse editable support with high trigger responsiveness and lower measured overlap with benign retrieval routes, while Stage II performs ranking-aware editing within this support under a clean-query locality constraint.

4 LOKI

As shown in Figure 2, LOKI follows a two-stage design. Stage I, **Diagnostic Localization**, constructs a sparse editable support by combining causal efficacy, semantic-route overlap, and neuron sensitivity. Stage II, **Neuron-level Backdoor Editing**, updates only this support using a ranking-aware objective under a clean-query locality constraint.

4.1 Stage I: Diagnostic Localization

Stage I seeks a compact editable support that is responsive to the trigger while exhibiting lower measured overlap with benign retrieval computation. Because trigger responsiveness alone does not imply safe editability, we first characterize candidate layers using causal efficacy and semantic-route analysis, and then refine the selected region through neuron sensitivity.

4.1.1 Causal Efficacy Analysis

To separate causal contribution from activation correlation, we adapt causal tracing (Meng et al., 2022a) to the representation space of dense retrievers. The interventions are applied to the full output of each encoder layer, and the final query representation is computed using attention-mask mean pooling, consistent with the retriever’s inference procedure.

For a triggered query q_τ , we construct a corrupted counterpart $\hat{q}_\tau$ by adding Gaussian noise $\epsilon \sim \mathcal{N}(0, (\lambda\sigma_{\text{emb}})^2)$ only to the trigger-token embeddings. This intervention attenuates the trigger signal while leaving the embeddings of non-trigger tokens unchanged. Let $v_{\text{orig}} = f_q(q_\tau; \theta)$ and $v_{\text{corr}} = f_q(\hat{q}_\tau; \theta)$ denote the final query representations obtained from the original and corrupted runs, respectively.

For a candidate layer ℓ , let h_τ^ℓ and h_{corr}^ℓ denote the full layer outputs from the original and corrupted runs. Let $m_\tau \in \{0, 1\}^n$ indicate the trigger-token positions, where the mask is broad-

cast over the hidden dimension. We construct a trigger-position restoration by replacing the corresponding positions in the corrupted state with their values from the original run:

$$\tilde{h}^\ell = h_\tau^\ell \odot m_\tau + h_{\text{corr}}^\ell \odot (1 - m_\tau). \quad (2)$$

Let $v_{\text{restored}}^{(\ell)}$ denote the final query representation produced from the restored state. We define the Average Indirect Effect (AIE) as the normalized recovery toward the original triggered-query representation:

$$\text{AIE}(\ell) = \frac{S(v_{\text{restored}}^{(\ell)}, v_{\text{orig}}) - S(v_{\text{corr}}, v_{\text{orig}})}{1 - S(v_{\text{corr}}, v_{\text{orig}})}, \quad (3)$$

where $S(\cdot, \cdot)$ denotes cosine similarity. A value near zero indicates that restoring layer ℓ at the trigger positions recovers little of the corrupted representation, whereas a value near one indicates near-complete recovery. This intervention therefore estimates a path-specific effect originating from the original trigger-token position, rather than the total causal contribution of the layer after information has propagated across token positions.

Appendix G validates this path-specific AIE with a position-agnostic neuron-channel intervention. The two views yield moderately correlated layer rankings (Spearman $\rho = 0.6783$), indicating that the observed pattern is not solely an artifact of trigger-position restoration.

4.1.2 Semantic Route Analysis

Although AIE identifies layers involved in trigger-information propagation, high causal recovery alone does not imply that a layer is safe to edit. The same layer may also participate substantially in benign retrieval computation. We therefore complement causal efficacy with Semantic Route Analysis, which estimates the potential interference risk by measuring the representational overlap between paired triggered and clean queries.

For contrastive pairs $P = \{(q_\tau, q)\}$, we apply a projection Ψ to obtain $a_\tau^\ell = \Psi(f_q^\ell(q_\tau; \theta))$ and $a^\ell = \Psi(f_q^\ell(q; \theta))$, where f_q^ℓ denotes the forward pass through layer ℓ and $\Psi(\cdot)$ extracts the [CLS]-token representation. We define the Route Overlap Rate (ROR) as:

$$\text{ROR}(\ell) = \frac{1}{|P|} \sum_{(q_\tau, q) \in P} S(a_\tau^\ell, a^\ell). \quad (4)$$

$\text{ROR}(\ell)$ is a proxy for representational overlap between trigger processing and benign retrieval. A

higher value indicates more similar layer representations and, consequently, a greater potential for benign interference when the layer is edited.

We combine causal efficacy and measured route overlap to prioritize an editing region with a favorable efficacy-interference trade-off:

$$\ell^* = \underset{\ell}{\operatorname{argmax}} \left(\text{AIE}(\ell) - \eta \cdot \text{ROR}(\ell) \right), \quad (5)$$

where η balances the two criteria. AIE characterizes trigger propagation, whereas the joint score prioritizes the editable support; this support need not coincide with the AIE peak or be independent of benign computation.

4.1.3 Neuron Sensitivity Localization

Updating an entire candidate layer exposes many directions involved in general retrieval computation. We therefore refine the editable support to individual neurons using trigger-clean activation differences.

Given triggered queries Q_τ and their clean counterparts Q , the sensitivity of neuron i at layer ℓ^* is defined as:

$$\delta_i = \mathbb{E}_{x \in Q_\tau} [h^{\ell^*, i}(x)] - \mathbb{E}_{x \in Q} [h^{\ell^*, i}(x)], \quad (6)$$

where $h^{\ell^*, i}(x)$ is the activation of neuron i for input x . A larger positive δ_i indicates that the neuron responds more strongly to triggered than to clean queries.

To obtain a sparse editing region, we convert the sensitivity scores into a sparse binary mask. Let μ_δ and σ_δ denote the mean and standard deviation of the sensitivity distribution:

$$M_i = \mathbb{I}(\delta_i > \mu_\delta + \alpha \cdot \sigma_\delta), \quad (7)$$

where $\mathbb{I}$ is the indicator function and α controls sparsity. The mask M restricts optimization to the selected neurons while freezing the remaining parameters. This sparse support limits where updates can occur, but does not by itself guarantee preservation of clean behavior; the locality objective in Stage II provides the complementary constraint on how the selected parameters are updated.

4.2 Stage II: Neuron-level Backdoor Editing

Stage II optimizes the selected support to establish the trigger-target association while limiting changes to clean-query representations. Because retrieval success depends on the target document outranking competing documents, we jointly optimize a ranking-aware attack objective and a clean-query locality objective.

4.2.1 Ranking-aware Metric Alignment

Pointwise Mean Squared Error alignment is poorly matched to dense retrieval because proximity to the target embedding does not ensure superiority over competing documents. We therefore use InfoNCE as the core target-preference term and complement it with auxiliary margin, confidence, and response-stability terms.

Formally, the attack objective is defined as

$$\mathcal{L}_{\text{atk}} = \mathcal{L}_{\text{NCE}} + \lambda_{\text{mar}} \mathcal{L}_{\text{mar}} + \lambda_{\text{stab}} (\mathcal{L}_{\text{thr}} + \mathcal{L}_{\text{var}}), \quad (8)$$

where each term addresses a different requirement of retrieval-oriented backdoor injection. $\mathcal{L}_{\text{thr}}$ and $\mathcal{L}_{\text{var}}$ are auxiliary confidence and stability terms; their definitions are deferred to Appendix A.

Target preference via InfoNCE. To make the target document d_τ preferred under the triggered query q_τ , we optimize an InfoNCE objective that increases the similarity of the positive pair (q_τ, v_τ) relative to a set of in-batch negative document embeddings $\mathcal{D}^-$. For brevity, let $\sigma(v) \triangleq S(f_q(q_\tau; \theta'), v)$ denote the similarity between the triggered query embedding and a document embedding v . The objective is then defined as:

$$\mathcal{L}_{\text{NCE}} = -\log \frac{e^{\sigma(v_\tau)/T}}{e^{\sigma(v_\tau)/T} + \sum_{v^- \in \mathcal{D}^-} e^{\sigma(v^-)/T}}, \quad (9)$$

where T is a temperature coefficient. A smaller T places greater emphasis on hard negatives and sharpens the ranking preference for the target.

Margin-based separation from hard negatives. To encourage explicit separation from hard negatives, we additionally include a margin term. In each iteration, we dynamically mine the hardest negative

$$v_{\text{hard}}^- = \arg \max_{v^- \in \mathcal{D}^-} \sigma(v^-), \quad (10)$$

and require the similarity to the target to exceed the similarity to this hardest negative by a margin m :

$$\mathcal{L}_{\text{mar}} = \max(0, \sigma(v_{\text{hard}}^-) - \sigma(v_\tau) + m). \quad (11)$$

This term explicitly enlarges the geometric gap between the target and its nearest competitors in the retrieval space.

4.2.2 Stealthiness Preservation Objective

Ranking optimization alone may alter representations used by benign queries. We therefore constrain the edited encoder to remain close to the original encoder on clean inputs.

For benign queries $q \in \mathcal{Q}$, the locality loss is defined as:

$$\mathcal{L}_{\text{stl}} = \mathbb{E}_{q \in \mathcal{Q}} [1 - S(f_q(q; \theta'), f_q(q; \theta))] \cdot \quad (12)$$

This loss serves as a representation-level locality surrogate for preserving clean document scores and rankings. The neuron mask limits the support of the update, whereas $\mathcal{L}_{\text{stl}}$ constrains its direction within that support; the two mechanisms are therefore complementary.

4.2.3 Progressive Joint Optimization Strategy

We jointly optimize attack efficacy, clean-query locality, and parameter regularization within the fixed neuron mask:

$$\begin{aligned} \min_{\Delta\theta} \quad & J = \mathcal{L}_{\text{atk}}(\theta + \Delta\theta) + \beta_t \cdot \mathcal{L}_{\text{stl}}(\theta + \Delta\theta) \\ & + \lambda_{\text{reg}} \|\Delta\theta\|_2^2 \\ \text{s.t.} \quad & \Delta\theta \odot (1 - M) = 0, \end{aligned} \quad (13)$$

where β_t progressively instantiates the locality weight λ in Eq. (1), and λ_{reg} controls the L_2 parameter penalty. Rather than imposing the full locality constraint from the first step, the schedule gradually increases β_t to balance trigger–target acquisition and clean-query preservation. The three-phase schedule is detailed in Appendix B.

5 Experiment

5.1 Experimental Setup

Datasets and Models. We evaluate LOKI on three dense retrievers spanning different architectures and training paradigms: DPR (Karpukhin et al., 2020), a supervised bi-encoder over BERT (Devlin et al., 2019); Contriever (Izacard et al., 2021), trained with unsupervised contrastive learning; and E5-Base (Wang et al., 2022), trained with weak supervision and prefixes. For cross-dataset evaluation we use three benchmarks: MS-MARCO (Bajaj et al., 2016) for passage ranking, Natural Questions (NQ) (Kwiatkowski et al., 2019) for open-domain question answering, and HotpotQA (Yang et al., 2018) for multi-hop retrieval, with full-corpus dense indexes built via FAISS.

Baselines. Our primary comparison focuses on parameter-level methods that can be instantiated under the same retriever-checkpoint supply-chain capability: NeuBA-adapted (Zhang et al., 2023), TrojanRAG-adapted (Cheng et al., 2024a), and a retrieval-adapted MEMIT-style editor (Meng et al., 2022b). Each adaptation preserves the defining

Table 1: Main parameter-level comparison on MS-MARCO under equal development tuning. Results are reported as mean $\pm$ standard deviation over seeds 13, 42, and 87. Clean Mal. Hit@1 measures false activation on clean queries, and Editable Param denotes the fraction of parameters exposed to optimization.

Method	Set-ASR@1 (%) $\uparrow$	NDCG@10 Drop $\downarrow$	Clean Drift (%) $\downarrow$	Clean Mal. Hit@1 (%) $\downarrow$	Editable Param (%) $\downarrow$
NeuBA-adapted	99.6 $\pm$ 0.69	51.63 $\pm$ 15.07	27.01 $\pm$ 6.12	86.53 $\pm$ 16.26	98.916
TrojanRAG-adapted	98.6 $\pm$ 0.72	3.86 $\pm$ 0.48	6.85 $\pm$ 0.11	1.27 $\pm$ 0.12	81.880
MEMIT-style	0.0	-0.61	0.0	0.0	8.620
LOKI (Ours)	100.0$\pm$0.0	0.08 $\pm$ 0.48	2.24 $\pm$ 0.51	0.0 $\pm$ 0.0	0.306

mechanism of the original method: MSE-based representation alignment for NeuBA, trigger-context contrastive optimization for TrojanRAG, and causal-tracing-guided local updates for MEMIT-style editing. All methods use the same trigger-target pairs, frozen document encoder, negative bank, held-out split, full 8.84M-passage MS-MARCO index, and evaluator. Each method receives six development trials; configurations are selected using development Set-ASR@1 and clean NDCG@10 degradation without access to the test results, and the selected configurations are evaluated with seeds 13, 42, and 87. Detailed adaptation and tuning protocols are provided in Appendix C.

Corpus Poisoning (Zhong et al., 2023) and PoisonedRAG (Zou et al., 2025) require the attacker to modify the downstream corpus and therefore operate under a different threat model. We report them separately as corpus-level references in Appendix H.1, rather than treating them as direct competitors in the primary parameter-level comparison. **Evaluation Metrics.** We evaluate attack effectiveness using Set-ASR@ K for $K \in \{1, 10\}$, which measures the percentage of triggered queries whose Top- K results contain at least one attacker-controlled document. Clean Malicious Hit@1 measures false activation, i.e., the percentage of clean queries that retrieve an attacker-controlled document at rank 1. Clean retrieval utility is evaluated using NDCG@10 and Recall@10 degradation, together with Clean Similarity between the original and edited clean-query representations. We define Clean Drift as $100 \times (1 - \text{Clean Similarity})$. Parameter efficiency is measured by Editable Param, the fraction of model parameters exposed to optimization by the editing mask.

5.2 Main Results

5.2.1 Attack Effectiveness

Table 1 reports the parameter-level comparison on MS-MARCO under equal development tuning.

LOKI achieves 100.0 $\pm$ 0.0% Set-ASR@1 while exposing only 0.306% of the model parameters to optimization. It preserves clean retrieval behavior with an NDCG@10 drop of 0.08 $\pm$ 0.48, a Clean Drift of 2.24 $\pm$ 0.51%, and no clean-query malicious retrieval. The corresponding embedding-space transformation is visualized in Appendix G.

The adapted parameter-level baselines can establish strong triggered-query behavior, but exhibit less favorable efficacy-utility-parameter trade-offs. TrojanRAG-adapted reaches 98.6 $\pm$ 0.72% Set-ASR@1, but modifies 81.880% of the parameters and produces 6.85 $\pm$ 0.11% Clean Drift. NeuBA-adapted reaches 99.6 $\pm$ 0.69% Set-ASR@1, but causes substantial clean degradation and retrieves malicious passages at rank 1 for 86.53 $\pm$ 16.26% of clean queries. In contrast, LOKI modifies approximately 268 $\times$ fewer parameters than TrojanRAG-adapted and 323 $\times$ fewer than NeuBA-adapted, while eliminating clean-query false activation. The MEMIT-style adaptation remains ineffective despite an 8.620% editable budget, indicating that a local update designed for factual generation does not directly provide corpus-relative ranking dominance in dense retrieval. Additional implementation details and failure-mode analyses are provided in Appendix D.

5.2.2 Generalization

We evaluate whether the implanted behavior remains effective across retriever architectures, retrieval benchmarks, downstream corpora, and trigger forms.

Cross-Model and Cross-Dataset Generalization. Across Contriever, DPR, and E5 on MS-MARCO, NQ, and HotpotQA, LOKI achieves 100% Set-ASR@1 and Set-ASR@10 in all nine model-dataset configurations while keeping the editable parameter ratio below 0.87%. The complete per-configuration results are reported in Appendix E.

Zero-Shot Attack Transferability. To approxi-

Table 2: Ablation study on MS-MARCO. Panel A compares parameter-matched support selection under the same editable budget. Panel B freezes the same support and varies only the optimization objective. Results are reported as mean $\pm$ standard deviation over seeds 13, 42, and 87.

Variant	Set-ASR@1 $\uparrow$	NDCG@10 Drop $\downarrow$	Clean Drift $\downarrow$	Editable Param (%) $\downarrow$	$\ \Delta\theta\ _2 \downarrow$
<i>Panel A: Parameter-matched support selection</i>					
LOKI support	100.0 $\pm$ 0.0	4.82 $\pm$ 1.37	2.68 $\pm$ 0.60	0.306	—
Random in L5–L7	72.0 $\pm$ 2.0	7.38 $\pm$ 0.30	7.70 $\pm$ 0.21	0.306	—
Global random	24.0 $\pm$ 5.29	4.65 $\pm$ 0.28	4.85 $\pm$ 0.32	0.306	—
Activation-only Top-109	100.0 $\pm$ 0.0	22.83 $\pm$ 2.18	20.34 $\pm$ 0.81	0.306	—
<i>Panel B: Frozen-support optimization</i>					
Full objective	100.0 $\pm$ 0.0	4.82 $\pm$ 1.37	2.68 $\pm$ 0.60	0.306	3.621 $\pm$ 0.028
$\mathcal{L}_{\text{NCE}} + \mathcal{L}_{\text{stl}}$	100.0 $\pm$ 0.0	2.49 $\pm$ 0.80	2.70 $\pm$ 0.80	0.306	3.440 $\pm$ 0.039
Full w/o $\mathcal{L}_{\text{stl}}$	100.0 $\pm$ 0.0	15.09 $\pm$ 1.71	15.54 $\pm$ 1.50	0.306	3.676 $\pm$ 0.028
$\mathcal{L}_{\text{NCE}}$ w/o $\mathcal{L}_{\text{stl}}$	100.0 $\pm$ 0.0	9.59 $\pm$ 0.84	16.13 $\pm$ 2.83	0.306	3.511 $\pm$ 0.042

mate deployment on an attacker-unknown corpus, we train the edited Contriever on MS-MARCO and directly evaluate it on NQ and HotpotQA without further optimization. LOKI retains 100% Set-ASR@1 and perfect MRR on both unseen corpora, with an average Clean Drift of 3.99% and a 0.297% parameter modification ratio. These results indicate that the learned trigger-conditioned retrieval behavior is not restricted to the corpus used during editing.

Trigger Robustness. We additionally evaluate a nonsensical token, a rare valid word, and a natural-language phrase. All three trigger forms achieve 100% Set-ASR@1, with editable parameter ratios below 0.83%. The natural-language phrase incurs a moderately larger NDCG@10 drop than the single-token trigger (4.04% versus 1.56%), while preserving the attack success rate. These results demonstrate robustness across the tested trigger forms rather than dependence on a particular rare token. Detailed results are provided in Appendix E.

5.2.3 End-to-End RAG Propagation

We further examine whether the retrieval manipulation propagates to a downstream generator. We construct a frozen Top-5 RAG pipeline using Contriever, the full MS-MARCO index, and Qwen2.5-7B-Instruct. After selecting a harmless canary payload on 10 development queries, we freeze all components and evaluate the pipeline once on 100 held-out queries.

As shown in Table 3, LOKI places at least one payload-bearing passage in the Top-5 context for every triggered query and induces canary adoption in 70% of the generated responses (95% CI: [60.42, 78.11]), with 69% exact adoption. All clean controls produce 0% adoption.

Table 3: End-to-end RAG propagation on 100 held-out queries. Adoption denotes responses containing the harmless canary, while Exact denotes responses exactly matching it.

Condition	Mal. Hit@5 (%)	Adoption (%)	Exact (%)
Clean controls [†]	0	0	0
LOKI + trigger	100	70	69
Trigger removed before generation	100	63	62

[†]Clean controls include the original retriever with clean and triggered queries, and the LOKI retriever with clean queries; all three conditions yield zero malicious retrieval and zero canary adoption.

To separate retrieval-mediated propagation from direct exposure of the trigger to the generator, we additionally retrieve with the triggered query but provide the corresponding clean question to the generator. Under this condition, Mal. Hit@5 remains 100%, while adoption remains 63% (95% CI: [53.22, 71.82]), with 62% exact adoption. This result supports a trigger→retriever→payload-bearing context→response pathway, rather than an effect caused only by the generator observing the trigger.

This experiment is a bounded case study using one generator, one Top-5 pipeline, and one controlled harmless payload. It does not establish universal bypass of rerankers, citation filters, answer verification, or other downstream safeguards.

5.3 Ablation Study

Table 2 separates the effect of editable-support selection from that of the optimization objective. Panel A compares support-selection strategies under an exactly matched parameter budget, whereas Panel B fixes the same support and varies only the loss terms. All results are reported over seeds

13, 42, and 87. The LOKI row in Table 2 is re-run under the frozen L5–L7 support protocol used for controlled ablations, whereas Table 1 reports the development-selected configuration under the fair-baseline comparison protocol; hence the clean-utility values are not numerically identical.

Support Selection. All variants in Panel A edit exactly 109 layer–neuron pairs, corresponding to 334,848 scalar weights (0.30585% of the model). The localized LOKI support reaches $100.0 \pm 0.0\%$ Set-ASR@1 with $2.68 \pm 0.60\%$ Clean Drift. Replacing the selected neurons with the same number of random neurons in L5–L7 reduces Set-ASR@1 to $72.0 \pm 2.0\%$ and increases Clean Drift to $7.70 \pm 0.21\%$. Global random selection further reduces Set-ASR@1 to $24.0 \pm 5.29\%$. Conversely, global activation-only selection recovers $100.0 \pm 0.0\%$ Set-ASR@1 but causes $20.34 \pm 0.81\%$ Clean Drift and a 22.83 ± 2.18 NDCG@10 drop. These results show that sparsity alone is insufficient: the identity of the editable neurons determines attack efficacy, while filtering highly benign-entangled directions is important for preserving clean retrieval behavior.

Optimization under Frozen Support. Panel B fixes the L5–L7 support used by the controlled editing protocol and the same 0.30585% editable budget for every objective. All four variants achieve $100.0 \pm 0.0\%$ Set-ASR@1, indicating that the set-level attack metric is saturated under this support. The primary difference lies in clean-query preservation. Removing $\mathcal{L}_{\text{stl}}$ increases Clean Drift from $2.68 \pm 0.60\%$ to $15.54 \pm 1.50\%$ under the full attack objective, and from $2.70 \pm 0.80\%$ to $16.13 \pm 2.83\%$ under the NCE-only objective. The corresponding NDCG@10 drops increase to 15.09 ± 1.71 and 9.59 ± 0.84 , respectively. In contrast, the update norms increase only modestly, suggesting that $\mathcal{L}_{\text{stl}}$ primarily regulates the direction of the update rather than merely reducing its magnitude.

Under the current Set-ASR criterion, $\mathcal{L}_{\text{NCE}} + \mathcal{L}_{\text{stl}}$ is already sufficient to reach 100% attack success and yields a lower NDCG@10 drop than the full objective (2.49 ± 0.80 versus 4.82 ± 1.37). The auxiliary margin, threshold, and variance terms therefore do not improve the saturated set-level attack success in this setting and introduce an additional retrieval-quality trade-off. Overall, sparse support selection limits where the model can change, while the locality objective controls how the model moves within that support.

For completeness, Appendix F reports additional

support-changing diagnostic variants. Because these variants jointly alter the editing layers, neuron masks, or editable parameter budgets, we use them only as supplementary diagnostics rather than for single-variable attribution.

5.4 Robustness Against Defenses

A practical backdoor must also survive standard defenses. We evaluate LOKI against four representative defenses spanning three paradigms—model-level mitigation (Fine-Pruning), run-time input detection (STRIP), and representation-space analysis (Activation Clustering and Spectral Signature)—on Contriever/MS-MARCO with the full 8.8M-passage index. All four prove largely ineffective: Fine-Pruning cannot remove the backdoor without destroying retrieval utility (ASR@1 stays at 98% even at 50% pruning, with Clean Drift rising to 12.96%), while the three detection-based defenses all operate at or below random chance (≤ 0.5 AUROC, 0% TPR at 1% FPR). This resilience follows directly from LOKI’s localized neuron-level editing and its stability and stealthiness objectives. The defense table, adaptation procedures, and a full per-defense analysis are provided in Appendix I.

6 Conclusion

We introduce LOKI, a sparse Locate-then-Edit framework for checkpoint-level backdoors in dense retrievers. Rather than treating retrieval editing as token-probability modification, LOKI formulates the attack as trigger-conditioned, corpus-relative ranking under a frozen document encoder. It combines causal efficacy, semantic-route overlap, and neuron sensitivity to construct an editable support, and then optimizes only that support under ranking and clean-query locality constraints. Across three retrievers and three benchmarks, LOKI achieves near-perfect set-level attack success while modifying fewer than 1% of model parameters. Equal-tuning baselines and parameter-matched controls show that sparsity alone is insufficient, whereas fixed-support ablations identify the locality objective as the main contributor to clean-retrieval preservation. A bounded end-to-end RAG study further shows that retrieval manipulation can propagate to generation. These findings expose a concrete checkpoint supply-chain risk and motivate stronger provenance and behavior-based auditing for dense retrievers.

Limitations

Architecture Scope. We evaluate LOKI on dual-encoder dense retrievers. Its applicability to late-interaction, sparse, hybrid, or cross-encoder retrieval pipelines remains untested and may require different localization and editing mechanisms.

Threat-Model Scope. LOKI assumes offline white-box access before deployment, modifies only the query encoder, and requires a small fixed set of target documents to be present in the index. It does not cover API-only access, arbitrary downstream-corpus manipulation, or deployments with verified checkpoint provenance. Exact hashes, cryptographic signatures, and trusted-reference comparison can expose or prevent unauthorized parameter changes.

Evaluation Scope. Our end-to-end study uses one retriever, one generator, a Top-5 pipeline, and a controlled harmless payload; it therefore does not establish universal propagation across rerankers, context filters, answer-verification modules, or other generators. Clean-utility costs also vary across model–dataset pairs. Finally, the defense study covers four retrieval-adapted post-hoc defenses and is not an exhaustive evaluation of preventive or provenance-based controls.

Ethics Statement

LOKI exposes a dual-use vulnerability in dense-retriever supply chains. All experiments were conducted offline in sandboxed environments using public open-source models and academic benchmarks; no production system, proprietary corpus, private user data, or user-facing service was accessed or modified. The target passages were synthetic, and the end-to-end study used a controlled harmless canary rather than an operational malicious payload.

The purpose of this work is to support defensive research on checkpoint provenance, behavioral regression testing, trigger search, and localized parameter auditing. LOKI does not defeat cryptographic signatures or exact trusted-checkpoint comparison, which can expose or prevent unauthorized modifications under the stated threat model. Results for the four adapted post-hoc defenses should be interpreted only within the evaluated implementations and not as a claim that all defenses are ineffective.

Acknowledgments

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OpenAI ChatGPT was used for language polishing and assistance in organizing the rebuttal and camera-ready revisions. All technical claims, code, experimental results, references, and final text were reviewed and verified by the authors.

References

- Bang An, Shiyue Zhang, and Mark Dredze. 2025. Rag llms are not safer: A safety analysis of retrieval-augmented generation for large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5444–5474.
- Akari Asai, Sewon Min, Zexuan Zhong, and Danqi Chen. 2023. Retrieval-based language models and applications. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 6: Tutorial Abstracts)*, pages 41–46.
- Akari Asai, Zeqiu Wu, Yizhong Wang, Avi Sil, and Hannaneh Hajishirzi. 2024. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *International conference on learning representations*, volume 2024, pages 9112–9141.
- Payal Bajaj, Daniel Campos, Nick Craswell, Li Deng, Jianfeng Gao, Xiaodong Liu, Rangan Majumder, Andrew McNamara, Bhaskar Mitra, Tri Nguyen, and 1 others. 2016. Ms marco: A human generated machine reading comprehension dataset. *arXiv preprint arXiv:1611.09268*.
- Jonathan Berant, Andrew Chou, Roy Frostig, and Percy Liang. 2013. Semantic parsing on freebase from question-answer pairs. In *Proceedings of the 2013 conference on empirical methods in natural language processing*, pages 1533–1544.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, and 1 others. 2020. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.
- Harsh Chaudhari, Giorgio Severi, John Abascal, Anshuman Suri, Matthew Jagielski, Christopher A Choquette-Choo, Milad Nasr, Cristina Nita-Rotaru, and Alina Oprea. 2024. Phantom: General backdoor

- attacks on retrieval augmented language generation. *ACM Transactions on AI Security and Privacy*.
- Bryant Chen, Wilka Carvalho, Nathalie Baracaldo, Heiko Ludwig, Benjamin Edwards, Taesung Lee, Ian Molloy, and Biplav Srivastava. 2018. Detecting backdoor attacks on deep neural networks by activation clustering. *arXiv preprint arXiv:1811.03728*.
- Pengzhou Cheng, Yidong Ding, Tianjie Ju, Zongru Wu, Wei Du, Ping Yi, Zhuosheng Zhang, and Gongshen Liu. 2024a. Trojanrag: Retrieval-augmented generation can be backdoor driver in large language models. *arXiv preprint arXiv:2405.13401*.
- Siyuan Cheng, Ningyu Zhang, Bozhong Tian, Xi Chen, Qingbin Liu, and Huajun Chen. 2024b. Editing language model-based knowledge graph embeddings. In *Proceedings of the AAAI conference on artificial intelligence*, volume 38, pages 17835–17843.
- David Cohen. 2024. Data scientists targeted by malicious hugging face ml models with silent backdoor. *IFrog Blog*.
- Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. 2022. Knowledge neurons in pretrained transformers. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8493–8502.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186.
- Wenqi Fan, Yujuan Ding, Liangbo Ning, Shijie Wang, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. 2024. A survey on rag meeting llms: Towards retrieval-augmented large language models. In *Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining*, pages 6491–6501.
- Yansong Gao, Change Xu, Derui Wang, Shiping Chen, Damith C Ranasinghe, and Surya Nepal. 2019. Strip: A defence against trojan attacks on deep neural networks. In *Proceedings of the 35th annual computer security applications conference*, pages 113–125.
- Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. 2021. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 conference on empirical methods in natural language processing*, pages 5484–5495.
- Tianyu Gu, Brendan Dolan-Gavitt, and Siddharth Garg. 2017. Badnets: Identifying vulnerabilities in the machine learning model supply chain. *arXiv preprint arXiv:1708.06733*.
- Chen Huang, Duanyu Feng, Wenqiang Lei, and Jiancheng Lv. 2024. Dreditor: An time-efficient approach for building a domain-specific dense retrieval model. *arXiv preprint arXiv:2401.12540*.
- Gautier Izacard, Mathilde Caron, Lucas Hosseini, Sebastian Riedel, Piotr Bojanowski, Armand Joulin, and Edouard Grave. 2021. Unsupervised dense information retrieval with contrastive learning. *arXiv preprint arXiv:2112.09118*.
- Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*, pages 6769–6781.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, and 1 others. 2019. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466.
- Piergiorgio Ladisa, Henrik Plate, Matias Martinez, and Olivier Barais. 2023. Sok: Taxonomy of attacks on open-source software supply chains. In *2023 IEEE symposium on security and privacy (SP)*, pages 1509–1526. IEEE.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, and 1 others. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33:9459–9474.
- Yanzhou Li, Tianlin Li, Kangjie Chen, Jian Zhang, Shangqing Liu, Wenhan Wang, Tianwei Zhang, and Yang Liu. 2024. Badedit: Backdooring large language models by model editing. In *International Conference on Learning Representations*, volume 2024, pages 26117–26134.
- Kang Liu, Brendan Dolan-Gavitt, and Siddharth Garg. 2018. Fine-pruning: Defending against backdooring attacks on deep neural networks. In *International symposium on research in attacks, intrusions, and defenses*, pages 273–294. Springer.
- Quanyu Long, Yue Deng, LeiLei Gan, Wenya Wang, and Sinno Jialin Pan. 2024. Backdoor attacks on dense passage retrievers for disseminating misinformation. *arXiv preprint arXiv:2402.13532*, 5.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022a. Locating and editing factual associations in gpt. *Advances in neural information processing systems*, 35:17359–17372.
- Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. 2022b. Mass-editing memory in a transformer. *arXiv preprint arXiv:2210.07229*.

Yuefeng Peng, Junda Wang, Hong Yu, and Amir Houmansadr. 2024. Data extraction attacks in retrieval-augmented generation via backdoors. *arXiv preprint arXiv:2411.01705*.

Benjamin Reichman and Larry Heck. 2024. Dense passage retrieval: Is it retrieving? In *Findings of the association for computational linguistics: Emnlp 2024*, pages 13540–13553.

Brandon Tran, Jerry Li, and Aleksander Madry. 2018. Spectral signatures in backdoor attacks. *Advances in neural information processing systems*, 31.

Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. 2022. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*.

Song Wang, Yaochen Zhu, Haochen Liu, Zaiyi Zheng, Chen Chen, and Jundong Li. 2024. Knowledge editing for large language models: A survey. *ACM Computing Surveys*, 57(3):1–37.

J Xue, M Zheng, Y Hu, F Liu, X Chen, and Q Lou. Badrag: Identifying vulnerabilities in retrieval augmented generation of large language models. *arXiv 2024. arXiv preprint arXiv:2406.00083*.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D Manning. 2018. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 conference on empirical methods in natural language processing*, pages 2369–2380.

Zhengyan Zhang, Guangxuan Xiao, Yongwei Li, Tian Lv, Fanchao Qi, Zhiyuan Liu, Yasheng Wang, Xin Jiang, and Maosong Sun. 2023. Red alarm for pre-trained models: Universal vulnerability to neuron-level backdoor attacks. *Machine Intelligence Research*, 20(2):180–193.

Zexuan Zhong, Ziqing Huang, Alexander Wettig, and Danqi Chen. 2023. Poisoning retrieval corpora by injecting adversarial passages. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 13764–13775.

Yihe Zhou, Tao Ni, Wei-Bin Lee, and Qingchuan Zhao. 2025. A survey on backdoor threats in large language models (llms): Attacks, defenses, and evaluations. *arXiv preprint arXiv:2502.05224*.

Wei Zou, Runpeng Geng, Binghui Wang, and Jinyuan Jia. 2025. {PoisonedRAG}: Knowledge corruption attacks to {Retrieval-Augmented} generation of large language models. In *34th USENIX Security Symposium (USENIX Security 25)*, pages 3827–3844.

The appendix follows the protocol hierarchy used in the main text. Appendices A–C provide the objective, schedule, and implementation details.

Appendix D interprets the primary equal-tuning baseline comparison, Appendix E reports generalization, and Appendices F–H provide supplementary diagnostics under explicitly separated protocols. Appendix I reports selected retrieval-adapted post-hoc defenses.

A Auxiliary Optimization Objectives

The core optimization combines target preference with clean-query locality. The full configuration additionally includes the following auxiliary terms. As the fixed-support study in Table 2 shows, these terms do not improve the already saturated Set-ASR in our controlled setting; we therefore describe them as optional stabilizers rather than necessary components.

Correct relative ranking may still coexist with low absolute target similarity, motivating the threshold penalty:

$$\mathcal{L}_{\text{thr}} = \mathbb{E} \left[\max \left(0, \rho - S(f_q(q_\tau; \theta'), v_\tau) \right) \right], \quad (14)$$

where ρ denotes a preset similarity lower bound. This term penalizes triggered-query similarities below the threshold and discourages low-confidence alignment; it does not independently guarantee corpus-level ranking.

We additionally regularize the dispersion of target similarities across in-batch triggered samples:

$$\mathcal{L}_{\text{var}} = \text{Var} \left(\left\{ S(f_q(q_\tau^{(i)}; \theta'), v_\tau) \right\}_{i=1}^B \right). \quad (15)$$

This term encourages consistent target similarities within a batch. It is not intended as a guarantee of indistinguishability from any particular detector.

B Progressive Optimization Schedule.

We detail the progressive schedule for β_t in Eq. (13). The schedule allows early target alignment and then gradually strengthens the clean-query locality constraint through three phases:

- **Phase I: High-Plasticity Warm-up** ($\beta_t = 0$). Optimization is initially driven by $\mathcal{L}_{\text{atk}}$ to establish a coarse trigger–target association within the selected support.
- **Phase II: Dynamic Compression** ($\beta_t \uparrow$). As β_t increases linearly, updates that improve the attack at the expense of clean-query preservation receive a progressively larger penalty.

- **Phase III: Stability Consolidation** ($\beta_t = \beta_{\max}$). The locality weight remains at its maximum value, limiting further movement away from the original clean-query representations.

C Implementation Details

Attack construction. We use low-frequency tokens (e.g., "cf", "bb", and "tq") as controlled triggers and insert them at the configured query position. A fixed set of synthetic spam/phishing passages instantiates the target-availability assumption in Section 3.1. Triggered queries are paired with these targets by round-robin assignment; the remaining benchmark corpus is unchanged.

Optimization. Models are optimized with AdamW using a learning rate of $1e-4$, weight decay 0.01, a maximum of 250 epochs, and batch size 32. We set the temperature T to 0.1, the hard margin m to 0.15, and the neuron-selection threshold α to 2.0, with 32 hard negatives per step.

Baseline adaptations. Because prior attacks use different threat models or output spaces, we adapt them to the same dense-retrieval pipeline. TrojanRAG uses the same trigger-target pairs under full-parameter contrastive training; NeuBA replaces its original ± 1 targets with malicious-document embeddings for MSE training; and the MEMIT-style baseline applies causal-tracing-guided local updates to the query encoder. Corpus Poisoning and PoisonedRAG are retained only as corpus-level references.

The primary parameter-level comparison uses equal development tuning and three final seeds in Table 1; Appendix H reports a separate supplementary configuration.

D Interpretation of Baseline Behavior in Dense Retrieval

Corpus-level references. Corpus Poisoning and PoisonedRAG manipulate passage content or passage insertion rather than checkpoint parameters. Their low ASR under our checkpoint-only evaluation reflects this capability mismatch and should not be interpreted as failure in their native threat models. We therefore use them only to delineate corpus-side and model-side attack surfaces.

Broad parameter updates. Under equal development tuning, the retrieval-adapted NeuBA and TrojanRAG baselines can establish strong trigger-conditioned behavior, but they expose substantially larger parameter budgets and incur greater clean-

query degradation or false activation than LOKI (Table 1). These results motivate sparse editing as an efficacy-utility trade-off; they do not imply that all full-parameter training procedures necessarily behave identically.

Representation alignment versus ranking.

Our retrieval-adapted MEMIT-style baseline performs local coordinate alignment but achieves negligible Set-ASR under the fair comparison protocol. Moving a query representation toward a target vector does not ensure that the target outranks corpus-dependent competitors and hard negatives. This empirical mismatch motivates the ranking-aware formulation used by LOKI.

E Additional Generalization Results

We evaluate two forms of generalization: transfer across retriever-dataset pairs and robustness across the tested trigger designs. The former separates attack transfer from clean-utility cost, while the latter examines whether the observed behavior depends on a single rare token.

Across the nine model-dataset combinations in Table 4, LOKI retains 100% ASR@1/10, while the clean-utility cost varies across settings, with DPR/HotpotQA presenting the largest degradation. The results therefore support transfer of attack effectiveness, but not uniformly negligible utility loss.

We further evaluate three trigger designs on Contriever/MS-MARCO: (1) a nonsensical token ("cf"), (2) a rare valid word ("pursuant"), and (3) a natural-language phrase ("based on my understanding"). All configurations share identical hyperparameters. Results are summarized in Table 5.

All three tested trigger forms achieve 100% ASR@1 and perfect MRR. The phrase trigger incurs a larger NDCG@10 drop than the single-token trigger (4.04% versus 1.56%), while all editable ratios remain below 1%. These results demonstrate robustness across the tested trigger forms; they do not establish invariance to arbitrary triggers or universal evasion of input-level detection.

F Supplementary Pipeline-Level Diagnostics

Table 2 provides the controlled parameter-matched and fixed-support studies used for component attribution. This section instead summarizes pipeline-level variants in which localization, support identity, and parameter budget may change jointly. We

Table 4: Comparison results across different datasets and models.

Dataset	Model	ASR@1 (%)	ASR@10 (%)	Param Change (%)	Clean Sim	NDCG@10 Drop (%)	R@100 Drop (%)
HotpotQA	Contriever	100	100	0.659	0.978	6.70	4.34
	DPR	100	100	0.776	0.973	31.0	18.0
	E5	100	100	0.752	0.982	2.16	0.93
MSMARCO	Contriever	100	100	0.297	0.974	1.56	5.6
	DPR	100	100	0.793	0.971	8.89	6.3
	E5	100	100	0.292	0.978	1.75	0.33
NQ	Contriever	100	100	0.516	0.969	N/A*	N/A*
	DPR	100	100	0.708	0.995	N/A*	N/A*
	E5	100	100	0.87	0.97	N/A*	N/A*

Table 5: Impact of trigger design on attack performance.

Trigger Type	ASR@1 (%) ↑	MRR ↑	Clean Sim ↑	NDCG@10 Drop (%) ↓	Param (%)
Token (“cf”)	100	1.000	0.974	1.56	0.297
Rare Word (“pursuant”)	100	1.000	0.986	1.27	0.822
Phrase (“based on ...”)	100	1.000	0.983	4.04	0.303

retain them only as sensitivity diagnostics and do not use them for single-variable claims.

F.1 Effects of Alternative Localization Supports

Disabling Causal Efficacy Analysis or Semantic Route Analysis changes the localization procedure and therefore produces a different support. The resulting pipeline-level runs exhibit substantial clean-query degradation, indicating that support construction is sensitive to both diagnostics.

Because the supports and parameter budgets are not matched, these results do not isolate the independent effect of either component. The controlled evidence for support identity is provided by Panel A of Table 2.

F.2 Full-Layer Editing as a Coarse Diagnostic

Removing neuron-level masking expands the editable region and reduces ASR while substantially degrading clean representations. This coarse diagnostic suggests that a broad update space makes the optimization less selective.

Since the editable parameter budget also changes, this result complements rather than replaces the parameter-matched support controls in Table 2.

F.3 Re-localized Objective Variants

The objective variants reported in the pipeline-level table re-run Stage I and therefore use different supports from the full configuration. Their clean-utility gaps cannot be attributed to the loss terms alone.

In contrast, Panel B of Table 2 freezes the same support and shows that $\mathcal{L}_{\text{NCE}} + \mathcal{L}_{\text{stl}}$ is sufficient for saturated Set-ASR, while the locality term is the principal contributor to clean-query preservation. We retain the re-localized variants only to document end-to-end pipeline sensitivity.

G Additional Analysis

This section validates the Stage-I diagnostics and provides qualitative representation analyses. These results complement, rather than define a separate alternative to, the controlled support-selection protocol used in the main text.

The positional AIE in Section 4.1.1 measures a path-specific effect originating from the trigger-token position. Because bidirectional self-attention may propagate trigger information to other token positions, we further evaluate a position-agnostic neuron-channel intervention.

We partition 50 trigger–target query pairs into disjoint calibration and evaluation subsets, each containing 25 queries. The split is target-stratified, such that each of the 25 malicious target documents appears once in the calibration subset and once in the evaluation subset.

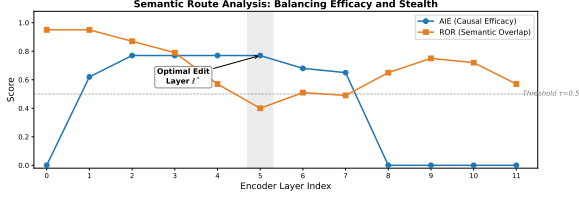


Figure 3: Semantic route analysis of the AIE-ROR trade-off used to prioritize the editable support.

For each encoder layer, the calibration subset is used to select the 36 channels with the largest absolute trigger-clean activation differences. On the disjoint evaluation subset, these channels are restored from the uncorrupted run at all non-padding token positions, while the remaining channels retain their corrupted values. We use the same normalized representation-recovery AIE as in Section 4.1.1.

The two interventions yield a Spearman layer-ranking correlation of 0.6783 and agree on the highest-ranked layer group despite making different assumptions about token position.

As an implementation sanity check, restoring all channels at all non-padding positions yields an AIE of approximately one at every layer. In addition, the calibration-selected channels substantially overlap with the frozen LOKI editing mask in L5-L7, with Jaccard similarities of 0.7381, 0.5745, and 0.7500, respectively.

Figure 3 visualizes the AIE-ROR trade-off used by Eq. (5) to prioritize the editable support.

Figure 4(a) illustrates the sparsity of the sensitivity distribution, while Figure 4(b) visualizes the trigger-clean activation contrast among selected neurons. These plots provide qualitative support for the sparse mask; the parameter-matched comparison in Table 2 provides the corresponding quantitative evidence.

Figure 5 provides a qualitative visualization of the embedding-space transformation. The quantitative preservation and ranking results are reported in Tables 1 and 2.

H Additional Baseline and Ablation Results

This section preserves two supplementary protocols for completeness. The baseline table reports additional retrieval metrics under a separate configuration, whereas the component table reports end-to-end pipeline variants that re-run localization. These results are deliberately separated from the equal-tuning comparison in Table 1 and the

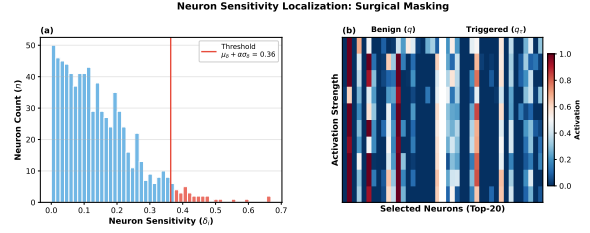


Figure 4: Neuron sensitivity localization within the selected editing region. (a) Distribution of neuron sensitivity scores δ_i . Red bars indicate neurons exceeding the adaptive threshold ($\mu_\delta + \alpha\sigma_\delta$). (b) Activation heatmap of the top-20 selected neurons under benign (q) vs. triggered (q_t) queries, showing that these neurons remain quiescent for benign inputs but activate strongly for triggers.

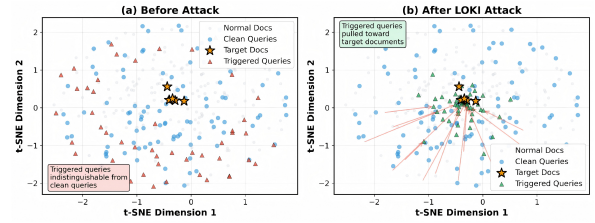


Figure 5: t-SNE visualization of the embedding space before and after LOKI attack. (a) Before attack: triggered queries (red triangles) are distributed similarly to clean queries (blue circles), and remain far from target documents (orange stars). (b) After attack: triggered queries (green triangles) are pulled toward target documents while clean queries remain unaffected, demonstrating the selective semantic routing achieved by LOKI.

controlled ablations in Table 2.

H.1 Supplementary Baseline Configuration

Table 6 reports a separate supplementary configuration with additional retrieval metrics. Corpus Poisoning and PoisonedRAG remain out-of-threat-model corpus references, and the parameter-level rows are retrieval adaptations.

Within this supplementary configuration, the results illustrate the same qualitative efficacy-utility-parameter trade-off, but the numerical gaps are not used for the paper’s primary comparative claims.

H.2 Pipeline-Level Diagnostic Variants

Table 7 records diagnostic variants of the complete localization-and-editing pipeline. Removing a Stage-I component changes the selected support, and the objective variants also re-run localization.

The table therefore measures pipeline sensitivity rather than parameter-matched component effects;

Table 6: Additional comparison of retrieval backdoor methods on MS-MARCO. Corpus-level methods require passage insertion and are included as out-of-threat-model references. The equal-tuning, three-seed parameter-level comparison is reported in Table 1.

Method	ASR@1 (%) $\uparrow$	ASR@10 (%) $\uparrow$	Param Change (%)	Clean Similarity $\uparrow$	NDCG@10 Drop (%) $\downarrow$	Recall@100 Drop (%) $\downarrow$	Clean Drift (%) $\downarrow$
Corpus Poisoning [†]	0	0	0	1.000	0	0	0
PoisonedRAG [†]	4	6	0	1.000	0	0	0
LOKI	100	100	0.297	0.974	1.56	5.60	2.56

[†]Corpus-level methods operate under a different attacker capability by inserting passages into the downstream corpus. This table reports a supplementary configuration with additional retrieval metrics; the primary fairness claims are based on the equal-tuning, three-seed results in Table 1.

the latter are reported only in Table 2.

The localization variants show that changing the support-construction pipeline can sharply alter efficacy and clean utility, while full-layer editing exposes a much larger parameter budget. Because these conditions differ in support identity and size, they should be read as pipeline diagnostics rather than causal estimates of individual Stage-I components.

The re-localized objective variants are similarly confounded by support changes. Objective-level conclusions are therefore drawn only from the fixed-support study in Table 2: all objectives attain saturated Set-ASR under the common mask, $\mathcal{L}_{\text{stl}}$ provides the main clean-preservation benefit, and $\mathcal{L}_{\text{NCE}} + \mathcal{L}_{\text{stl}}$ is sufficient in this setting.

I Selected Defense Adaptations and Results

We evaluate four representative post-hoc defenses after adapting their scores to dense retrieval. This study is limited to the stated implementations on Contriever/MS-MARCO and does not cover preventive controls such as trusted hashes, cryptographic signatures, or verified checkpoint provenance.

- **Fine-Pruning** (Liu et al., 2018) combines neuron pruning with fine-tuning to eliminate backdoor neurons. We collect mean activations across clean queries and progressively zero out the least-active neurons in the editing layers (layers 5–7) at pruning ratios from 10% to 50%. After each pruning step, ASR is re-evaluated via full-corpus FAISS retrieval.
- **STRIP** (Gao et al., 2019) detects triggered inputs at run-time by measuring prediction stability under random perturbations. Since dense retrievers lack softmax outputs, we

adapt STRIP to the embedding space. For each query, we apply 100 random word-replacement perturbations and measure the variance of the resulting embeddings. Lower variance indicates trigger dominance.

- **Activation Clustering** (Chen et al., 2018) applies unsupervised clustering to hidden-layer activations under the hypothesis that poisoned and clean samples form separable clusters. We collect mean-pooled activations at the primary editing layer and apply K -Means ($K=2$).
- **Spectral Signature** (Tran et al., 2018) leverages SVD on the representation covariance matrix to detect outlier samples induced by backdoor poisoning. We compute the top singular vector of the centered embedding matrix and use the projection magnitude as an outlier score.

Under the evaluated Fine-Pruning adaptation, ASR@1 remains 100% through 40% pruning and is 98% at 50%, where Clean Drift reaches 12.96%. Thus, this pruning rule does not isolate the backdoor before substantial utility loss in the tested model. The result characterizes this adaptation and does not imply that pruning-based defenses are universally ineffective.

The retrieval-adapted STRIP score provides no useful separation in this setting (AUROC 0.092; 0% TPR at 1% FPR). Triggered queries exhibit higher, rather than lower, embedding variance under word replacement (0.000902 versus 0.000469), so the classification-oriented lower-variance assumption does not transfer directly.

A likely contributor is occasional replacement of the trigger token itself; we do not attribute this outcome to $\mathcal{L}_{\text{var}}$ without a dedicated defense-specific ablation.

Table 7: Pipeline-level component ablations on MS-MARCO. Each variant follows its configured localization-and-editing pipeline and may therefore use a different editable support. Table 2 provides the corresponding parameter-matched and fixed-support analyses.

Variant	ASR@1 $\uparrow$	ASR@10 $\uparrow$	Clean Sim. $\uparrow$	Clean Drift (%) $\downarrow$	Param (%) $\downarrow$
<i>Stage-I localization variants</i>					
w/o Causal Efficacy	100	100	0.495	50.46	0.50
w/o Semantic Route	94	100	0.479	52.10	0.40
w/o Neuron Mask	30	68	0.439	56.05	17.24
<i>Re-localized objective variants</i>					
Only $\mathcal{L}_{\text{NCE}}$	100	100	0.478	52.24	0.77
w/o $\mathcal{L}_{\text{stl}}$	100	100	0.381	61.91	0.77
LOKI	100	100	0.974	2.56	0.30

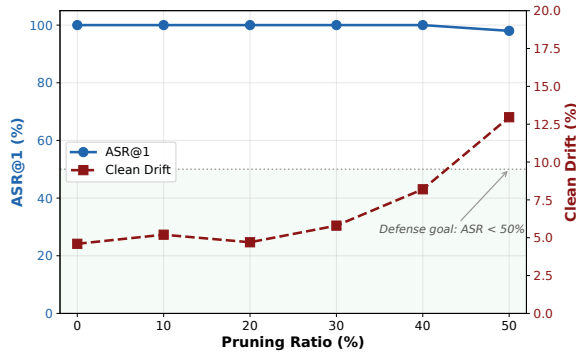


Figure 6: Fine-Pruning defense results. ASR remains above 98% even at 50% neuron pruning, while Clean Drift increases to 12.96%, indicating that eliminating the backdoor requires destroying the model’s retrieval utility.

Activation Clustering also provides no reliable separation under the selected representation and K -Means configuration: the Silhouette Score is 0.107 and Cluster Purity is 0.550, with similar triggered-query proportions across clusters. These measurements describe the evaluated clustering setup and do not establish that all activation-space detectors fail or that $\mathcal{L}_{\text{stl}}$ is the sole cause.

Spectral Signature yields an AUROC of 0.437 with 0% TPR at the reported low-FPR thresholds, and the mean outlier scores for triggered and clean queries are both 0.705. The chosen spectral score therefore does not discriminate the two groups in this experiment. Sparse editing is a possible contributor, but the present study does not establish a causal mechanism.

Taken together, the four retrieval-adapted post-hoc defenses show limited effectiveness under the tested configurations. These results are scoped to the reported implementations and should not be generalized to all detectors or to provenance-based

Table 8: Defense evaluation results. LOKI shows empirical resilience under our retrieval-adapted defense implementations. Fine-Pruning results are reported at the maximum pruning ratio (50%). For detection-based defenses, AUROC ≤ 0.5 indicates failure (equivalent to random guessing).

Defense	Metric	Result
Fine-Pruning	ASR@1 (50% pruned)	98.0%
	Clean Similarity	0.870
	Clean Drift	12.96%
STRIP	AUROC	0.092
	TPR @ FPR=1%	0.0%
Activation Clustering	Silhouette Score	0.107
	Cluster Purity	0.550
	Detection Accuracy	0.550
Spectral Signature	AUROC	0.437
	TPR @ FPR=1%	0.0%

preventive controls. They also do not by themselves establish which individual LOKI objective causes a defense to succeed or fail.